Foundry Inventory — Material & Inventory Operations

Scientific Identity, Physical Accounting & Operational Correctness

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***Abstract*—This specification applies Foundry Operations Algebra — Foundations to material and inventory operations. It derives scientific identity, physical containment, accountable quantities, decisions, and operational eligibility from the foundational primitives and state model. Versioned MaterialSpecifications and ContainerSpecifications define identity and compatibility boundaries; provenance and composition remain distinct from available stock. Immutable, balanced material movements establish quantity correctness and distinguish historical correction from physical reconciliation. Concrete task and workflow applications connect scientific and physical evidence to authorized effects, including partial failure and uncertain execution. Twelve detailed conformance cases and implementation-mapping obligations support evaluation at both foundational and material-domain levels. The material-accounting constraints supplement, rather than redefine or weaken, the foundational guarantees.**

***Index Terms*—Material identity, Inventory accounting, Scientific provenance, Containment, Eligibility, Laboratory workflows, Conformance**

I. Scope and Algebraic Foundation

This specification derives the material and inventory specialization of Foundry Operations Algebra — Foundations, Section I, v1.7 (hereafter **Foundations**). It covers scientific identity, containment, quantities, provenance, composition, material decisions, operational eligibility, and the laboratory operations that change them.

The foundational algebra defines the general semantics and correctness requirements for modeling laboratory operations. Domain models specialize its primitives, state structures, operations, and invariants. A domain specialization may introduce additional constraints but must not weaken or contradict the foundational guarantees.

A. Responsibilities and Limits

The material domain owns its scientific identities, containment representation, and quantity accounting. Decision state has an explicitly designated decision authority; task and workflow execution retain their own lifecycle authority. These logical responsibilities do not require separate services or a single shared database.

Instrument control, scheduling, reservations, procurement and suppliers, specialized scientific processing, and comprehensive analytical-result management are outside this specification. They interact through defined references, evidence, effects, and dependencies. Customer experiments and service orders may supply work translated into internal tasks and workflows.

Balanced material postings are an additional requirement of this specialization, not a universal rule imposed on every domain by the foundations.

B. Derivation and Conformance

A design is evaluated at two levels: foundational conformance to Foundations, Section VIII, and material-domain conformance to Section VII and the twelve cases in Section IX. Passing one level does not establish the other.

The following mapping applies the derivation method (Foundations, Section VII); it does not add universal primitives.

General role	Material specialization
Entity	Material, Container, Location, ContainerContent; stable records for movements and decisions
Relationship	Placement, content-to-material/container links, derivation, composition, supersession
Specification	MaterialSpecification, ContainerSpecification, LocationSpecification, and applicable task or decision contracts
Task	TaskSchemas such as AliquotSample and ReviewMaterialQC, invoked as TaskExecutions
Event	Transfer confirmations, observations, checkpoint results, and execution transitions
Effect	Identity or content creation, placement changes, committed movement postings, and decision changes

A TaskSchema is a versioned operation contract: typed inputs and outputs, resource requirements and preconditions, permitted effect families, and a mandatory lifecycle with states, transitions, checkpoints, and CommitRules. A TaskExecution is one invocation, recording the bound schema version, actual inputs and outputs, actors, lifecycle history, checkpoint evidence, and proposed and committed effect references. A schema defines intended work; an execution records what happened (Foundations, Sections IV–V).

CommitRules govern when an execution may propose effects and which commitments are required for completion. Only the domain responsible for the affected state validates and commits those effects, checking evidence, authority, and invariants against current authoritative state. An instrument

success response or execution status cannot substitute for required commitments.

ContainerContent is an identifiable holding, not a second scientific Material. A committed MaterialMovement is an immutable record of an accepted quantity Effect, not evidence by itself that a physical action occurred. Decisions have their own scope and authority even when they govern material use.

II. Material Domain State

The material domain specializes Foundations, Section III.B into three connected representations:

$$S_M = (P, C, Q)$$

Here P is scientific identity, composition, and provenance; C is physical containment and locations; Q is recorded quantities and movement history. For the material-focused examples:

$$S = (S_M, D, T)$$

D contains recorded decisions, restrictions, and authorizations; T retains the task and workflow definitions, executions, and history defined in Foundations, Section IV. Here $S = (S_M, D, T)$ denotes a material-focused projection of the foundational state $\left((S_k)_{k\in K}, T\right)$: it retains relevant material and decision components together with execution state and omits unrelated domain components. It is not an alternative general state model or an exhaustive platform model. A quality decision may change D without changing S_M; an observation may refine P without moving material; an execution transition may change T without changing quantities.

A. Scientific Identity, Composition, and Provenance

Requirements for P are realized by Material and MaterialSpecification (Section III.A–Section III.B), provenance (Section IV.B), and composition (Section IV.C). Identity may survive division between containers, while transformation or independent production may require a new identity. Matching attributes alone do not decide identity. Component contributions must not duplicate available inventory (Section IV.D).

B. Physical Containment

Requirements for C are realized by Container, ContainerSpecification, Location, LocationSpecification, Placement, and ContainerContent (Section III.C–Section III.F). Specifications distinguish direct material holdings, child containers, fixed positions, and movable contents. Placement and occupancy constraints apply independently of scientific identity.

C. Material Quantities and Movements

Requirements for Q are realized by MaterialMovement and the movement algebra. Quantity changes must be balanced within compatible scopes, traceable, and auditable (Section V.I). Double-entry principles provide internal correctness without requiring financial terminology in laboratory-facing operations.

D. Operational Eligibility

Eligibility is derived for a particular intended use rather than stored as an intrinsic property of Material.

$$\text{Eligible}(m, u, t, S, \Sigma) \rightarrow \{\text{Yes}, \text{No}, \text{Unknown}\}$$

Where:

- m: Material or ContainerContent.
- u: Intended use.
- t: Assessment time.
- S: Relevant operational state.
- Σ: Applicable specifications and policies.

Eligibility may depend on quality decisions, observations, material specifications, quantities, containment, methods, and policy. Unknown is not approval. A TaskExecution requiring eligible material validates that condition at the appropriate lifecycle boundary; stale approval cannot bypass commitment-time validation.

Location requirements are checked against the holding's current placement and applicable LocationSpecification versions (Section III.E.a). For a use requiring BSL-2 containment, known placement outside an authorized BSL-2 scope makes Eligible return No; missing placement or required containment evidence yields Unknown, not approval. This is a use-specific policy requirement, not a universal rule for all living cells.

Workflow-Scoped Material Eligibility. Material eligibility is evaluated in two operational contexts, not as two independent eligibility states. Both use the existing Eligible(m,u,t,S,Σ) assessment; the workflow contract governs when it is checked and how changes are handled.

Pre-flight eligibility determines whether specified material holdings may be admitted to a workflow for its intended operations. The admission contract validates applicable quantity, quality, stability, containment, scientific evidence, and restrictions, and retains the material identities, governing Specification versions, and evidence or decision references supporting authorization.

In-flight continuation authorization determines whether an admitted workflow may initiate a subsequent operation. The applicable TaskSchema defines its required checkpoints, relevant condition changes, and intervention rules. Pre-flight authorization does not unconditionally authorize all future operations.

Workflow admission may establish material allocation or custody through the responsible workflow authority. Such allocation is distinct from physical consumption and does not itself create a MaterialMovement. Material availability and eligibility remain governed by their respective authoritative states.

Eligibility governs authorization for use, not the existence or recorded quantity of material. Available quantity is derived for a holding, intended use, and assessment time from recorded inventory, eligibility, and applicable restrictions; it is not a second independent balance (Section V.A).

a) *Stability and Provenance-Dependent Expiration:*

Expiration is a stability assessment, not necessarily an intrinsic, fixed Material property. Applicable MaterialSpecifications and validated evidence may distinguish source stability, transformations, preparation or opening time, hold-

ing-specific handling, storage conditions, and intended use. Aliquoting may preserve or shorten a limit; dilution, mixing, or transformation may require a distinct assessment. These are policy-dependent possibilities, not universal scientific rules.

$$\text{ExpirationAssessment}(m, c, u, t, S, \Sigma) \rightarrow \{\text{Valid}, \text{Expired}, \text{Unknown}\}$$

Here m is Material, c is its ContainerContent, u is intended use, t is assessment time, S includes provenance and operational history, and Σ supplies the applicable Specification and policy versions. Valid means the applicable stability criteria are satisfied; Expired means a stability limit has been exceeded; Unknown means required information cannot establish the outcome.

For uses requiring stability approval of c, Expired makes that holding ineligible and Unknown precludes approval. A holding-specific result does not automatically classify other holdings of the same Material. Valid satisfies only the stability condition, not other eligibility requirements. Time alone can change the derived assessment without a state write. Assessment, expiration, or restriction does not consume or dispose of material and does not change recorded quantity.

The declared storage condition is taken from the nearest enclosing Location that declares one, following the holding's Container and Location placement chain; assessment over time uses effective-dated placement and LocationSpecification bindings plus required condition evidence, not merely the current location or its nominal temperature. Missing required history or evidence yields Unknown; the governing stability policy defines clock start, qualifying temperature ranges, excursions, and any reset rules.

An accepted expiration assertion records its cutoff or stability criteria, Material/holding and use scope, applicable versions, source assertions, method and condition evidence, authority, and effective history. Assignment or amendment is a domain-validated scientific or decision Effect; prior assertions remain traceable rather than being overwritten.

Policies declare whether source validity matters at preparation, at later use, or both, and which provenance dependencies or condition changes require reassessment. Required unresolved reassessment cannot silently preserve approval; applicable restrictions govern use pending resolution. Reassessment may retain or revise a limit, but provenance alone neither copies dates to all descendants nor universally resets a stability period. Section IX.K exercises these distinctions.

III. Material Entities and Specifications

These entities and versioned Specifications specialize the foundational primitives. Scientific identity and physical holding identity remain distinct.

A. Material

Material represents an identifiable scientific entity, such as a protein, reagent, sample, or preparation.

A Material has stable identity, scientific metadata, an applicable MaterialSpecification version, and optional composition and provenance relationships.

One Material may occupy multiple ContainerContents.

We do not initially impose a universal hierarchy of material definitions, lots, and samples.

B. MaterialSpecification

MaterialSpecification specializes Specification.

It defines:

- Identity-defining scientific distinctions.
- Descriptive attributes.
- Validation requirements.
- Relevant provenance and production boundaries.

Two independently produced protein batches may require separate identities even if their measured attributes match.

An aliquot may preserve identity despite occupying a new physical container.

We define specification-relative identity:

$$m_1 \underset{\Sigma}{\equiv\!\equiv} m_2$$

For a proposed material output o:

$$\text{IdentityDecision}(\Sigma, o) \rightarrow \{\text{Reuse}(M), \text{Create}(M')\}$$

Identity validation may require scientific evidence or authorized judgment.

Materials must satisfy their applicable MaterialSpecification's identity and validation rules. Specification changes do not silently rewrite existing identities. Amendments preserve prior scientific assertions, supporting evidence, and applicable specification versions.

C. Container

Container represents a physical object capable of holding material, other containers, or both.

Examples include vials, tubes, plate wells, plates, boxes, and racks.

These share one generalized Container abstraction, distinguished by specifications. Container identity remains stable through relocation and lifecycle changes. Retirement requires resolution of active contents and child placements; it does not erase their history.

D. ContainerSpecification

A ContainerSpecification defines physical capabilities and constraints.

Example	Direct material	Child containers	Placement behavior
Vial	Yes	No	Movable
Tube	Yes	No	Movable
Plate well	Yes	No	Fixed to plate
Plate	No	Yes	Fixed well grid
Box	No	Yes	Movable container grid
Rack	No	Yes	Compatible slots

Specifications may define occupancy, compatibility, positions, geometry, capacity, and lifecycle rules.

E. Location

Location represents a physical place or address, such as a laboratory, room, freezer, or shelf.

Locations are distinct from movable Containers.

Locations may be nested and may contain Containers. Each Location references a versioned LocationSpecification, with effective-dated bindings and placement history retained. The enclosing-location path must be unambiguous and acyclic.

a) *LocationSpecification:*

LocationSpecification specializes Specification and defines a Location's declared operating envelope:

- condition: optional storage-condition declaration, including temperature and optional humidity ranges, with units.
- containment_level: declared biosafety or other containment classification, its applicable standard, and authorized scope.
- capabilities: supported process kinds, such as characterization.
- capacity: optional occupancy limits with an explicit measurement basis.
- environmental: additional applicable constraints, such as atmosphere, light exposure, or cleanliness.

Declarations describe requirements or qualified capabilities, not proof of actual conditions or current authorization. Applicable evidence and restrictions must establish their validity for the intended use. Scope and inheritance of containment and capabilities are explicit in the Specification; the nearest-storage-condition rule does not automatically propagate process authorization. Placement must respect applicable compatibility and capacity constraints.

Version changes follow the foundational applicability and reassessment rules; they do not silently reinterpret historical storage or authorize in-flight work. Condition excursions and relevant placement or Specification changes trigger reassessment where required by the governing material or task policy.

F. ContainerContent

ContainerContent represents scientific Material occupying a particular Container.

It connects:

- Material identity.
- Physical Container.
- Accountable quantity.
- Content-specific metadata and history.

The same Material may appear in multiple ContainerContents. Each holding has a stable identifier distinct from its Material and Container identities. Its bindings identify the holding being accounted for: ordinary transfers debit the source and credit a destination holding rather than retarget the source holding's identity. Relocating the enclosing Container preserves holding identity.

Within one direct material-holding Container, at most one active ContainerContent exists for each Material identity. Separate holdings of the same Material require distinct Containers, including child Containers where applicable. Additional receipts of the same Material credit the existing active holding while retaining distinct movement and provenance references; different receipt histories or labels alone do not create parallel balances.

A Container may hold multiple Materials where permitted by its ContainerSpecification, with each holding independently identifiable and accountable. Physically combining portions into one undifferentiated holding does not leave independently withdrawable component balances: same-Material additions share one holding, and preparations managed as one quantity follow the composite accounting rule (Section IV.D).

ContainerContent is the ordinary quantity-tracking unit. Active content must reference a compatible, active Container. Closure requires zero recorded quantity and resolved physical outcomes. Zero quantity alone does not close a holding; closure is explicit under applicable lifecycle rules. Closed identifiers are not reused, and their movement history remains traceable.

A separate bulk or in-transit holding model is deferred until required.

G. MaterialMovement

MaterialMovement records quantity changes associated with laboratory activity or correction.

Examples include receiving, transferring, consuming, producing, disposing, and reconciling material.

Current recorded quantities derive from committed movements.

Committed movements are immutable and corrected through additional linked movements.

IV. Relationships

Placement, provenance, composition, and decisions connect the entities without conflating physical location, scientific identity, and available quantity. Scientific identity boundaries follow Section III.B; movement alone does not establish a new Material identity.

A. Placement and Position

Placement is represented by:

$$\mathrm{PlacedIn}(\mathrm{Child}, \mathrm{Parent}, \mathrm{Position})$$

Position is an attribute of the relationship, not a mandatory independent Entity.

Examples:

$$\mathrm{PlacedIn}(\mathrm{Well}_1, \mathrm{Plate}_1, \mathrm{A1})$$

$$\mathrm{PlacedIn}(\mathrm{Vial}_1, \mathrm{Box}_1, \mathrm{B3})$$

Placement must satisfy compatibility, fixed-position, exclusivity, and acyclicity constraints. No child may have conflicting active physical positions; containment must not form cycles. A fixed well retains its specified parent plate and position.

B. Scientific Provenance

Provenance uses typed relationships between Materials and generating executions. Each derived Material retains its source Material and generating execution references.

For example:

$$\mathrm{DerivedFrom}(M_2, M_1, E)$$

Aliquoting may preserve scientific identity.

Digestion may establish a distinct derived Material.

Physical movement history remains traceable even when identity is unchanged.

Scientific dependencies may connect specific versioned assertions and decisions, not only Material identities. Where a downstream assertion or authorization depends on source composition, concentration, identity, stability, or other scientific knowledge, the governing Specification defines which source changes require reassessment. These dependencies preserve their source versions and evidence references.

C. Scientific Composition

For composite Material M_C:

$$\text{Composition}(M_C) = \{(M_i, r_i)\}_{i=1}^{n}$$

Where r_i expresses a scientifically defined contribution with an explicit measurement basis.

Composition may describe constituent identities, proportions, concentrations, or other meaningful quantities. Constituent identities, contribution values, and their measurement basis remain traceable.

It does not itself create independently available inventory balances.

$$\text{Composition}(M_C) \not\Rightarrow Q(C_C, M_i) = r_i Q(C_C)$$

D. Composite Material Accounting Rule

When an operation creates a combined preparation managed as one accountable quantity:

1) Input quantities are accounted for by committed movements.
2) The output Material identity is established under its MaterialSpecification.
3) The output occupies one or more ContainerContents.
4) Component contributions are recorded as scientific composition or provenance.
5) Those contributions do not create duplicate available inventory balances.
6) Output quantity is established using appropriate measurement and evidence.

The output quantity is not universally required to equal the sum of the input quantities.

E. Quality Decisions and Restrictions

Quality decisions are distinct from physical material state.

They may apply to Materials, ContainerContents, groups, scientific results, or deliverables.

Decisions retain scope, outcome, authority, evidence, and effective history.

TaskExecutions such as ReviewMaterialQC and ReviewResultQC may produce authorized decisions.

Eligibility is derived from decisions and other applicable conditions.

V. Material Movement Algebra

Balanced postings specialize the foundational requirement for valid, auditable, domain-authorized transitions. Their balance scopes and correction rules apply to material quantities, not to every other domain's effects.

A. Quantity State

Let A be the set of accountable material positions.

Positions may represent:

- Active ContainerContents.
- External receipt sources.
- Consumption destinations.
- Disposal destinations.
- Discrepancies.
- Production sources.

Active ContainerContents are physical accountable holdings under the identity and uniqueness rules in Section III.F. Receipt, consumption, disposal, discrepancy, and production positions have accounting roles; they are not available physical stock.

Let $Q(a)$ be shorthand for $Q_{\text{recorded}}(a)$, the recorded balance at position a. Balance equations are evaluated within a fixed Material and compatible measurement-dimension scope; those arguments are suppressed in this notation.

For ordinary active ContainerContents:

$$Q(a) \geq 0$$

unless specifically authorized otherwise.

Available quantity is derived from recorded quantity, eligibility, relevant inventory state, and applicable restrictions for intended use u at time t. In general:

$$Q_{\text{recorded}}(a) \neq Q_{\text{available}}(a, u, t)$$

For a physical holding with a nonnegative recorded balance:

$$0 \leq Q_{\text{available}}(a, u, t) \leq Q_{\text{recorded}}(a)$$

A nonphysical position, closed holding, or an eligibility outcome of No or Unknown authorizes zero available quantity; Unknown remains an unresolved assessment, not proof of physical absence. A negative recorded balance has zero available quantity. Even when eligible, additional restrictions may reduce the available amount. Without additional restrictions, an eligible active physical holding makes its full nonnegative recorded balance available. Availability is not independently posted or adjusted; changes to recorded quantity still require committed movements.

In Conformance Case 6, placing the vial on hold leaves the recorded balance unchanged while availability for validated use becomes zero:

$$Q_{\text{recorded}}(C_1) = 50\,\mu L$$
$$Q_{\text{available}}(C_1, u_{\text{validated}}, t) = 0\,\mu L$$

B. Postings

Each posting has a stable identity unique within its MaterialMovement. Its accounting payload is:

$$p_i = (a_i, m_i, \delta q_i)$$

Where:

- a_i: Accountable position.
- m_i: Material identity.
- δq_i: Signed quantity change with units.

A movement identifier and stable posting index may identify an entry. Identical payloads do not imply identical

postings. Negative postings reduce a position's quantity; positive postings increase it.

C. MaterialMovement

A movement is a finite indexed family of uniquely identifiable postings:

$$\mathrm{MM} = (p_i)_{i=1}^{n}$$

Let $I_{m,d}$ select posting indices for Material m and measurement dimension d, and I_a select indices at position a within the balance's Material/dimension scope. Each index is counted once, even when accounting payloads coincide.

$$\sum_{i \in I_{m,d}} \delta q_i = 0$$

Quantities must be expressed in compatible units. The resulting balance is:

$$Q'(a) = Q(a) + \sum_{i \in I_a} \delta q_i$$

Movements commit only when relevant invariants hold. Parenthesized rows in the examples denote indexed entries with indices and identifiers omitted for readability, not deduplicated entries. An index identifies a posting; it does not prescribe an execution or commitment order.

D. Transfer

Transfer 20 µL of M_1 between Contents C_1 and C_2:

$$\begin{aligned}\mathrm{MM}_{\text{transfer}} = ((&C_1, M_1, -20\,\mu L),\\ &(C_2, M_1, +20\,\mu L))\end{aligned}$$

Both postings commit together.

E. Consumption and Disposal

Consumption and disposal use the same underlying transfer/posting mechanism.

For consumption:

$$\mathrm{ConsumeMaterial}(C, q) \equiv \mathrm{TransferMaterial}(C, \mathrm{Consumed}, q)$$

For disposal:

$$\mathrm{DiscardMaterial}(C, q) \equiv \mathrm{TransferMaterial}(C, \mathrm{Disposed}, q)$$

These equivalences describe their quantity-accounting effects.

Consumed and Disposed are internal accountable destinations, not necessarily physical containers.

Domain-specific operations may impose additional authorization, handling, and evidence requirements.

If material is moved into a tracked physical waste container instead, its destination may be another ContainerContent.

That physical holding remains separately traceable until the waste leaves tracked inventory.

F. Production and Scientific Transformation

For input Material M_i:

$$\begin{aligned}\mathrm{MM}_{\text{input},i} = ((&C_i, M_i, -q_i),\\ &(\mathrm{Consumed}, M_i, +q_i))\end{aligned}$$

For output Material M_O:

$$\begin{aligned}\mathrm{MM}_{\text{output}} = ((&\mathrm{Produced}, M_O, -q_O),\\ &(C_O, M_O, +q_O))\end{aligned}$$

Input and output movements balance independently within their Material identities.

Scientific provenance and composition are recorded separately.

There is no universal requirement that:

$$q_O = \sum_i q_i$$

G. Corrections and Reconciliation

Committed MaterialMovements are immutable.

We distinguish correcting erroneous history from reconciling inventory against physical observation.

a) *ReverseMovement:*

For:

$$\mathrm{MM} = ((a_i, m_i, \delta q_i))_{i=1}^{n}$$

The reversal is:

$$\mathrm{ReverseMovement}(\mathrm{MM}) = ((a_i, m_i, -\delta q_i))_{i=1}^{n}$$

It compensates an erroneous recorded movement. Each reversal posting has a new identity and links to the original posting it compensates. The correspondence preserves every original index, even when payloads coincide.

No new physical transfer is implied.

b) *ReplaceMovement:*

$$\mathrm{ReplaceMovement}(\mathrm{MM}_{\text{old}}, \mathrm{MM}_{\text{new}}) = \mathrm{ReverseMovement}(\mathrm{MM}_{\text{old}}) + \mathrm{MM}_{\text{new}}$$

The correction commits as one validated group. Here $+$ groups the reversal and replacement postings for joint commitment; it never merges entries with equal payloads.

The original remains in history and is explicitly superseded.

c) *ReconcileQuantity:*

$$\delta q = q_{\text{observed}} - q_{\text{recorded}}$$

For recorded 50 µL and observed 45 µL:

$$\begin{aligned}\mathrm{MM}_{\text{reconcile}} = ((&C_1, M_1, -5\,\mu L),\\ &(\mathrm{Discrepancy}, M_1, +5\,\mu L))\end{aligned}$$

The reconciliation references observation, evidence, and authorization.

It does not invent a historical cause.

H. Correction Integrity

Under a one-effective-reversal policy:

$$\mathrm{EffectiveReversalCount}(\mathrm{MM}) \leq 1$$

Replacement establishes:

$$\mathrm{Supersedes}(\mathrm{MM}_{\text{new}}, \mathrm{MM}_{\text{old}})$$

Corrections are validated as complete groups against current authoritative state, including legitimate subsequent activity. A reversal or replacement must not invalidate later valid movements, holdings, or references, nor silently erase or reinterpret legitimate subsequent activity. If that activity prevents a valid correction, reject the proposal or require explicit, evidence-backed, authorized resolution.

For example, a 20 µL transfer into an initially empty holding is followed by legitimate 15 µL consumption, leaving 5 µL. Replacing the transfer with 10 µL would debit another 10 µL and leave −5 µL. With no authorized negative-balance exception, reject that correction; do not delete the consumption or fabricate a receipt. Replacing it with 15 µL instead would leave zero and pass the quantity test, but all other invariants and evidence requirements still apply.

I. Movement Invariants

Invariant	Requirement
Balance	Postings balance within compatible Material and dimension scopes
Valid quantity	No unauthorized negative active balances
Posting identity	Every posting is identifiable; equal payloads remain separate accounting entries
Atomicity	All postings of one movement commit together
Idempotency	A movement cannot be committed twice
Traceability	Movements reference originating executions or events
Immutability	Committed history is preserved
Compatibility	Material identity and unit rules are respected
Authorization	Restricted movements and corrections require permission
Correction integrity	Corrections cannot conflict or duplicate
Evidence	Corrections and observations retain supporting evidence

Balanced postings alone do not establish scientific or physical validity.

VI. Domain Operations

Operation	Meaning
RegisterMaterial	Register scientific Material identity
ReceiveMaterial	Introduce material into tracked inventory
TransferMaterial	Move quantity between accountable positions
PlaceContent	Associate Material with a compatible Container
RelocateContainer	Change a Container's placement
ConsumeMaterial	Account for material used
ProduceMaterial	Account for material produced
DeriveMaterial	Record scientific lineage
RecordComposition	Record constituent identities and contributions
DiscardMaterial	Account for material disposed
RetireContainer	End a Container's active lifecycle
ReverseMovement	Compensate erroneous movement history
ReplaceMovement	Supersede an erroneous movement
ReconcileQuantity	Align recorded quantity with verified observation
RecordObservation	Record scientific or physical evidence
RecordDecision	Record authorized quality or disposition decisions

These operations specialize the general transition contract: they propose entity, relationship, lifecycle, decision, or movement effects. TaskSchema CommitRules govern permission to propose (Foundations, Section V.D); the owning domain validates and commits. Their semantic composition follows Foundations, Section VI.E.

A. Permitted Effect Families

These operation families identify what may change; each proposal is still subject to current-state validation, applicable Specifications, evidence, and authorization by the owning domain.

- RegisterMaterial proposes scientific identity creation and Specification binding. PlaceContent proposes a holding and its Material/Container links; it does not itself introduce quantity. RelocateContainer changes placement. RetireContainer changes lifecycle only after active contents and child placements are resolved.
- ReceiveMaterial, TransferMaterial, ConsumeMaterial, ProduceMaterial, and DiscardMaterial propose movement postings. ReverseMovement, ReplaceMovement, and ReconcileQuantity propose the linked correction or discrepancy postings defined in Section V.G, with evidence and authority rather than an implied physical action.
- DeriveMaterial and RecordComposition propose scientific relationships and contribution assertions. Neither independently creates available stock.
- RecordObservation records evidence without automatically changing quantities. RecordDecision proposes a scoped, authorized decision; it may change eligibility without moving material.

The list is an effect-family contract, not a complete scientific protocol for each operation. Concrete TaskSchemas add their typed bindings, lifecycle conditions, and required outcomes (Section VIII).

VII. Material Domain Invariants

These material-specific guarantees supplement the foundations. The index points to the defining rules and does not replace the movement invariant table. Each guarantee must be traced to implementation checks and conformance tests.

Guarantee	Defining sections
Identity stability	Section III.A, Section III.C
Scientific identity and validation	Section III.B
Identity amendments and evidence	Section III.B
Nonnegative active quantities	Section V.I
Balanced, atomic, auditable movements	Section V.I
No duplicate component inventory	Section IV.D

Guarantee	Defining sections
Traceable composition contributions	Section IV.C
Exclusive physical placement	Section IV.A
Acyclic containment	Section IV.A
Placement compatibility	Section III.D, Section III.E.a, Section IV.A
Versioned location conditions and use constraints	Section III.E, Section II.D, Section II.D.a, Section VIII.E
Fixed well parent and position	Section IV.A
Compatible, active content holdings	Section III.F
Holding identity and active uniqueness	Section III.F
Retirement resolves contents and children	Section III.C
Traceable scientific provenance	Section IV.B
Dependent Scientific Assertion Integrity	Section IV.B, invariant below; Section IX.L
Correction distinct from reconciliation	Section V.G, Section V.H
Authorized restricted operations	Section V.I, Section IV.E
Context-dependent eligibility	Section II.D
Workflow admission and continuation authorization	Section II.D, authorization boundaries below, Section VIII.E, Section IX.L
Traceable, policy-governed stability assessment	Section II.D.a
Derived availability, not a second balance	Section V.A

Cross-domain coordination follows Foundations, Section VI.D it does not weaken any participating domain's invariants.

Dependent Scientific Assertion Integrity. When an authoritative scientific assertion is corrected or superseded, dependent assertions and decisions must be evaluated for reassessment according to their applicable Specifications. Required unresolved reassessment must not silently preserve an authorization that depends on superseded evidence. From commitment of the relevant correction, affected approvals must not grant new material-use authorizations on stale evidence; delayed dependency evaluation must not create a stale-approval window. This applies through dependent assertions and decisions wherever their governing Specifications require it, without automatically invalidating every descendant. Historical assertions, committed movements, observations, and legitimate downstream operations remain traceable. Reassessment does not itself reverse physical activity or quantity postings.

Authorization Boundaries During Execution. A correction or supersession that requires reassessment prevents new affected material-use authorizations from relying on stale evidence from the time that correction becomes authoritative.

A physical operation already authorized and initiated before that point is not retroactively undone. Its actual outcome must be established and recorded under the applicable execution contract. Any subsequent affected operation must satisfy the required continuation conditions or obtain an appropriately authorized intervention decision before initiation.

An operator decision may permit continuation only within the exceptions and authority defined by the governing Specifications. Asynchronous notification or reassessment must not silently bypass a mandatory intervention boundary. At that boundary, authorization must be checked against the relevant authoritative restrictions and evidence versions, or an equivalent enforceable barrier; notification delivery alone is insufficient.

A. Inherited Foundational Guarantees

Mandatory task lifecycle, checkpoint validation, owning-domain commitment, defined completion conditions, failure integrity, retry safety, supported atomic groups, and ordered dependencies remain obligations of Foundations, Section VIII. They are not redefined here or relaxed by a balanced movement.

A conforming material implementation must therefore preserve the foundational execution and composition guarantees as well as the material constraints above.

VIII. Task and Workflow Applications

Laboratory operations instantiate the TaskSchema, TaskExecution, and WorkflowExecution semantics of Foundations, Section IV. An operation name in Section VI denotes a domain operation, not automatically a complete TaskSchema or instrument command.

Each concrete TaskSchema pins its versioned input/output contracts, resource requirements, allowed effect families, lifecycle, checkpoints, and CommitRules. The following application contracts identify the required distinctions; they do not replace method-specific scientific or authorization rules.

A. Material Transfer

A transfer schema binds a source ContainerContent, destination holding or compatible Container, Material identity, requested quantity with units, and actor or instrument reference. Preconditions include compatible identity and containment, valid quantity, applicable use restrictions, and authority.

Outcome verification establishes the actual transfer, not merely command acceptance. Required checkpoints distinguish physical confirmation from movement commitment and execution resolution. CommitRules permit only evidenced movements; a newly required ContainerContent and the associated transfer postings may form one supported atomic effect group.

Confirmed partial transfer is represented even if the requested transfer fails. Failure or cancellation does not reverse committed effects or prove that no physical action occurred. A timeout enters an explicit unresolved state: missing acknowledgment proves neither non-occurrence nor success. Further evidence, authorized observation, or reconciliation resolves the uncertainty; unverified outcomes are not posted as confirmed movements.

Required movement commitments precede completion. Repeated commitment requests for the same logical effect must not duplicate postings. Repeating an instrument command may instead move more material: a physical retry requires known conditions permitted by the lifecycle contract or explicit authorization as a distinct action (Foundations, Section V.G). Cases 1, 2, and 9 in Section IX exercise these obligations.

B. Composite Preparation

A preparation schema binds input holdings and quantities, an output MaterialSpecification and compatible Container, method and resource references, and required scientific evidence. Its permitted families include consumption, output production, derivation, and composition.

Checkpoints establish actual input consumption, the output's scientific identity and quantity, and required provenance and contribution assertions. CommitRules distinguish confirmed consumed inputs from output quantities that have not yet been established. The composite accounting rule (Section IV.D) prevents duplicate constituent inventory.

The schema defines which effects are required for completion. Confirmed consumption survives later failure; neither atomic recorded effects nor workflow failure can restore physically consumed inputs. Cases 3 and 8 test these semantics.

C. Material Quality Review

A review schema binds a Material or ContainerContent, applicable policy version, intended decision scope, reviewer authority, and supporting observations. Outputs include the scoped decision and its evidence references.

A decision-authority checkpoint precedes proposal, and the designated domain commits the decision before required completion. This changes decision state, not quantity or placement. Eligibility is then assessed for the intended use and time under Section II.D; an unresolved assessment is not approval. Case 6 tests this separation.

D. Composition Applications

Content creation and quantity transfer may be composed atomically within a supported Material-domain transaction boundary:

$$\mathrm{PlaceContent}(M, V) \otimes \mathrm{TransferMaterial}(C_{\mathrm{source}}, C_{\mathrm{new}}, q)$$

Here the transfer references the content created in the same effect group. The Material domain may commit content creation and its transfer postings together.

This is atomic recorded commitment under Foundations, Section VI.B, not atomic physical execution. The transfer references the content created in the same group.

Disposal followed by retirement is an ordered dependency under Foundations, Section VI.C. Retirement requires disposal to be accounted for and all other retirement preconditions to hold. If retirement fails after disposal is committed, disposal remains authoritative; case 10 tests both this sequence and the supported atomic alternative.

Dependencies spanning authorities or transaction boundaries are coordinated through task or workflow execution as required by Foundations, Section VI.D, without assuming rollback of physical consequences.

E. Workflow Admission and Active Intervention

A workflow requiring material eligibility defines an admission contract and applicable continuation checkpoints through its versioned TaskSchemas.

A TaskSchema may require execution in a specified Location, such as the characterization cell, with a declared process capability and containment level. Admission and continuation checkpoints validate the actual execution location, applicable LocationSpecification version, required evidence, and current restrictions. A location name or capability declaration alone does not satisfy the precondition; relocation or relevant changes require validation at the applicable authorization boundary.

Admission binds the intended material inputs, their eligible use scopes, supporting scientific assertions and decisions, and applicable Specification versions. The coordinating workflow authority may establish exclusive allocation or custody under a separate operational contract. Admission and allocation do not imply that material has been physically consumed.

During execution, the workflow evaluates newly applicable restrictions and evidence changes at the intervention boundaries required by its TaskSchemas. An affected operation may begin only when the applicable continuation conditions are satisfied or an appropriately authorized operator has resolved the intervention.

Operator intervention records its scope, evidence, authority, decision, and relationship to the affected TaskExecution. Permitted outcomes may include continuation, additional verification, rework, quarantine, or termination. No operator decision overrides a non-waivable Specification requirement.

An operation already in physical execution may complete, partially complete, or remain unresolved. Its confirmed effects remain authoritative and are not automatically reversed. Required intervention prevents subsequent affected operations from being newly authorized; it does not assume instantaneous cancellation of physical actions.

The workflow must preserve the provenance of admission, interventions, physical outcomes, and resulting decisions. Allocation, instrument control, and notification mechanisms remain outside the material domain except through their defined contracts.

IX. Conformance Cases

The twelve detailed cases below specialize the foundational obligations rather than replace them. Their scientific identities, holdings, quantities, decisions, and corrections are evaluated under this material model; execution and commitment semantics remain governed by the foundations.

A. Conformance Example #1 — Aliquoting

a) *Natural Language:*

A scientist divides 100 µL of protein sample into two 50 µL aliquots in separate empty vials.

Both retain the same Material identity.

b) *Initial State:*

- Material M_1, governed by Σ_1.
- Source Vial V_1, Content C_1: 100 µL.
- Empty destination Vials V_2, V_3.

c) *Operations:*

$$\text{PlaceContent}(M_1, V_2) \to C_2$$
$$\text{PlaceContent}(M_1, V_3) \to C_3$$
$$\text{TransferMaterial}(C_1, C_2, 50\,\mu L)$$
$$\text{TransferMaterial}(C_1, C_3, 50\,\mu L)$$

The committed movement is:

$$\begin{aligned}\text{MM}_{\text{aliquot}} = ((&C_1, M_1, -100\,\mu L),\\ &(C_2, M_1, +50\,\mu L),\\ &(C_3, M_1, +50\,\mu L))\end{aligned}$$

d) *Expected Result:*

$$\text{Material}(C_1) = \text{Material}(C_2) = \text{Material}(C_3) = M_1$$
$$Q(C_1) = 0\,\mu L$$
$$Q(C_2) = 50\,\mu L$$
$$Q(C_3) = 50\,\mu L$$

Required physical confirmation and inventory commitment checkpoints are satisfied before completion.

Interruption after one aliquot must not imply completion or non-occurrence of the other.

B. Conformance Example #2 — Plate Well Transfer

a) *Natural Language:*

Transfer 20 µL of protein sample from a vial into well A1 of a 96-well plate.

b) *Initial State:*

- Material M_1.
- Source Vial V_1, Content C_1: 100 µL.
- Plate P_1 with 96-well specification.
- Well W_1 fixed at A1 and initially empty.

c) *Operations:*

$$\text{PlacedIn}(W_1, P_1, \text{A1})$$
$$\text{PlaceContent}(M_1, W_1) \to C_2$$
$$\text{TransferMaterial}(C_1, C_2, 20\,\mu L)$$

Movement:

$$\begin{aligned}\text{MM}_{\text{well}} = ((&C_1, M_1, -20\,\mu L),\\ &(C_2, M_1, +20\,\mu L))\end{aligned}$$

d) *Expected Result:*

- Source vial: 80 µL.
- Well A1: 20 µL.
- Same scientific Material.
- Fixed well position preserved.
- Required effects committed.

The content creation and transfer postings may form an atomic effect group within the Material domain.

C. Conformance Example #3 — Scientific Transformation

a) *Natural Language:*

Digest a protein sample using a reagent, producing a distinct scientific preparation.

b) *Operations:*

$$\text{RegisterMaterial}(M_2, \Sigma_2)$$
$$\text{ConsumeMaterial}\big(C_{\text{protein}}, q_1\big)$$
$$\text{ConsumeMaterial}\big(C_{\text{reagent}}, q_2\big)$$
$$\text{ProduceMaterial}\big(M_2, V_{\text{output}}, q_3\big)$$
$$\text{DeriveMaterial}\big(M_2, \{M_{\text{protein}}, M_{\text{reagent}}\}\big)$$

c) *Expected Result:*

- Distinct output Material identity.
- Input consumption recorded.
- Output quantity tracked.
- Provenance retained.
- No universal assumption of input/output quantity equality.
- Failed execution does not automatically restore consumed inputs.

D. Conformance Example #4 — Historical Correction

a) *Natural Language:*

The system recorded a 20 µL transfer, but the actual transfer was 15 µL.

b) *Algebra:*

$$\begin{aligned}\text{MM}_{\text{old}} = ((&C_A, M_1, -20\,\mu L),\\ &(C_B, M_1, +20\,\mu L))\end{aligned}$$
$$\begin{aligned}\text{MM}_{\text{new}} = ((&C_A, M_1, -15\,\mu L),\\ &(C_B, M_1, +15\,\mu L))\end{aligned}$$
$$\text{ReplaceMovement}(\text{MM}_{\text{old}}, \text{MM}_{\text{new}})$$

c) *Expected Result:*

- Net recorded transfer: 15 µL.
- Original history preserved.
- Correction auditable.
- No new physical transfer implied.

Reversal and replacement commit atomically within the Material domain.

E. Conformance Example #5 — Physical Reconciliation

a) *Natural Language:*

The system reports 50 µL in a vial, while verified observation establishes 45 µL.

b) *Algebra:*

$$\text{ReconcileQuantity}(C_1, 45\,\mu L, \text{Observation})$$
$$\begin{aligned}\text{MM}_{\text{reconcile}} = ((&C_1, M_1, -5\,\mu L),\\ &(\text{Discrepancy}, M_1, +5\,\mu L))\end{aligned}$$

c) *Expected Result:*

- Recorded quantity: 45 µL.
- Observation and authorization retained.
- Original history unchanged.
- Discrepancy traceable without inventing its cause.

F. Conformance Example #6 — Material QC Hold

a) *Natural Language:*

A reference standard vial contains 50 µL. A QC reviewer places it on hold for validated analytical methods.

b) *Algebra:*

Initially:

$$\text{Eligible}(C_1, u_{\text{validated}}, t, S, \Sigma) = \text{Yes}$$

The authorized decision is recorded:

$$\text{RecordDecision}(C_1, \text{Hold}, E_1)$$

$$D' = D \cup \{\text{Hold}(C_1, E_1)\}$$

$$S_{M'} = S_M$$

Afterward:

$$\text{Eligible}(C_1, u_{\text{validated}}, t', S', \Sigma) = \text{No}$$

c) *Expected Result:*

- Quantity and placement unchanged.
- Hold decision committed.
- Restricted use becomes ineligible.
- Other uses are evaluated under their policies.
- Required decision commitment precedes task completion.

G. Conformance Example #7 — Material Identity Boundary

a) *Natural Language:*

Two independently produced protein batches have identical measured sequence and concentration, but their specification requires distinct identities based on production origin.

b) *Algebra:*

$$\text{Attributes}(M_A) = \text{Attributes}(M_B)$$

Yet:

$$M_A \underset{\Sigma_1}{\not\equiv} M_B$$

Therefore:

$$\text{IdentityDecision}(\Sigma_1, \text{Batch}_B) \rightarrow \text{Create}(M_B)$$

c) *Expected Result:*

Separate Material identities and origin histories.

Matching attributes do not cause merging.

H. Conformance Example #8 — Composite Preparation

a) *Natural Language:*

Combine 80 µL of protein solution with 20 µL of buffer to produce one mixed preparation.

Assume additive volumes and no loss for this example.

b) *Operations:*

$$\text{RegisterMaterial}(M_C, \Sigma_C)$$

$$\text{ConsumeMaterial}(C_A, 80\,\mu L)$$

$$\text{ConsumeMaterial}(C_B, 20\,\mu L)$$

$$\text{ProduceMaterial}(M_C, V_C, 100\,\mu L)$$

$$\text{DeriveMaterial}(M_C, \{M_A, M_B\})$$

$$\text{RecordComposition}(M_C, \{(M_A, 0.8), (M_B, 0.2)\})$$

The input consumption movements are:

$$\text{MM}_A = ((C_A, M_A, -80\,\mu L), (\text{Consumed}, M_A, +80\,\mu L))$$

$$\text{MM}_B = ((C_B, M_B, -20\,\mu L), (\text{Consumed}, M_B, +20\,\mu L))$$

The output production movement is:

$$\text{MM}_C = ((\text{Produced}, M_C, -100\,\mu L), (C_C, M_C, +100\,\mu L))$$

c) *Expected Result:*

$$Q(C_C) = 100\,\mu L$$

$$\text{Composition}(M_C) = \{(M_A, 0.8), (M_B, 0.2)\}$$

One output quantity is available.

Its components are not counted again as independent inventory.

Confirmed consumption remains recorded if later preparation steps fail.

I. Conformance Example #9 — Automated Transfer Timeout

a) *Natural Language:*

An instrument receives a command to transfer 20 µL from a vial to a plate well.

Communication times out before completion is confirmed.

b) *Initial State:*

- Material M_1.
- Source Content C_A: 100 µL.
- Destination well empty.
- No committed transfer movement.

c) *Lifecycle Checkpoints:*

Checkpoint	Condition
CommandAccepted	Instrument acknowledges command
TransferConfirmed	Actual transferred quantity established
InventoryCommitted	Material domain confirms movement
ExecutionResolved	Required outcomes accounted for

After timeout:

$$\text{Running} \rightarrow \text{AwaitingEvidence}$$

No unverified movement is committed.

Blind physical retry is prohibited.

d) *Resolution A — Full Transfer:*

Evidence confirms 20 µL.

The Material domain commits:

$$\text{MM} = ((C_A, M_1, -20\,\mu L), (C_B, M_1, +20\,\mu L))$$

Result:

$$Q(C_A) = 80\,\mu L, \quad Q(C_B) = 20\,\mu L$$

e) *Resolution B — No Transfer:*

Reliable evidence confirms no movement.

Inventory remains unchanged.

The execution resolves according to its lifecycle contract.

f) *Resolution C — Partial Transfer:*

Evidence confirms only 12 µL moved.

The domain commits a 12 µL transfer.

The task may fail its intended 20 µL requirement without reversing the confirmed movement.

g) *Resolution D — Unknown Outcome:*

The outcome cannot be established.

No invented quantity is committed.

The execution and affected inventory remain subject to explicit resolution or reconciliation.

h) *Invalid Variations:*

- Timeout interpreted as proof of non-occurrence.
- Acknowledgment interpreted as proof of physical completion.
- Blind physical retries.
- Duplicate postings on repeated completion messages.
- Completion before required commitments.
- Automatic reversal of confirmed effects after failure.

J. Conformance Example #10 — Disposal and Container Retirement

a) *Natural Language:*

A vial contains 50 µL of material that must be disposed of. After the material is disposed of and accounted for, the vial is retired.

b) *Initial State:*

- Container V_1: Active vial.
- ContainerContent C_1: 50 µL of Material M_1.
- Disposal destination Disposed: Internal accountable destination excluded from available stock.

c) *Composition:*

The intended dependency may be expressed as:

$$\text{TransferMaterial}(C_1, \text{Disposed}, 50\,\mu L) \rightarrow \text{RetireContainer}(V_1)$$

This notation specifies that container retirement depends on the disposal being accounted for, under Foundations, Section VI.C.

d) *Disposal Movement:*

$$\text{MM}_{\text{disposal}} = ((C_1, M_1, -50\,\mu L), \\ (\text{Disposed}, M_1, +50\,\mu L))$$

e) *Expected Success:*

- Available material quantity becomes zero.
- Disposal is recorded with applicable evidence and authorization.
- Container retirement follows after its preconditions are satisfied.
- Both operations remain auditable.

f) *Failure After Disposal:*

Suppose disposal is confirmed and committed, but container retirement fails.

Expected behavior:

- Disposal remains committed.
- The vial remains active pending resolution.
- The coordinating task or workflow records the incomplete dependency.
- No automatic reversal of disposal occurs.

g) *Atomic Alternative:*

If the authoritative state changes for disposal and retirement are supported within one transaction boundary, their recorded effects may instead be grouped atomically:

$$\text{TransferMaterial}(C_1, \text{Disposed}, 50\,\mu L) \otimes \text{RetireContainer}(V_1)$$

This guarantees all-or-nothing commitment of those recorded effects.

It does not guarantee that physical disposal can be reversed.

h) *Invalid Variations:*

- Retiring a container with unresolved active contents.
- Treating an ordered dependency as automatically atomic.
- Reversing confirmed disposal because retirement failed.
- Treating database rollback as reversal of a physical disposal action.
- Duplicating disposal movements during recovery.

K. Conformance Example #11 — Provenance-Dependent Material Expiration

a) *Scenario and Policy Fixture:*

A scientist aliquots protein stock Material M_A, then uses another portion to prepare diluted Material M_B. All dates and rules below are illustrative fixture assumptions, not scientific recommendations. Cutoffs are explicit UTC instants; a holding is expired at or after its applicable cutoff.

- Stock holding C_S: 100 µL of M_A, with accepted source assertion A.expiry.v1 valid through 30 November 2026; its cutoff is 1 December 2026, 00:00 UTC.
- Buffer holding C_R: 25 µL of distinct Material M_R, with acceptable input stability and condition evidence.
- Empty, compatible destination vials for aliquot C_A and diluted preparation C_B; intended use u requires stability approval.
- Preparation time: 25 November 2026, 09:00 UTC. All required evidence is initially satisfactory.
- MaterialSpecification version A1 preserves scientific identity on aliquoting and limits a new aliquot to 48 hours after division, capped by the accepted source expiry.
- MaterialSpecification version B1 requires a distinct diluted identity, stable inputs at preparation, and validated method/handling evidence. It allows 24 hours after preparation and requires reassessment when referenced source stability assertions change.

b) *Preparation and Execution Contract:*

Transfer 50 µL from C_S to C_A without changing Material identity. Consume another 25 µL from C_S and 25 µL from C_R to produce 50 µL of M_B in C_B, assuming additive volumes and no loss. Commit the confirmed movements and retain source, method, and execution references.

Recorded quantities are then 25 µL in C_S, zero in C_R, and 50 µL in each output holding. Accepted assessments establish an aliquot cutoff of 27 November 2026, 09:00 UTC and a diluted-preparation cutoff of 26 November 2026, 09:00 UTC.

The preparation and reassessment TaskSchemas bind policy and assertion versions, inputs, actors, and evidence. Required checkpoints establish provenance and conditions, authorized assessment, and any required assertion or restriction commitment before completion. A resolved expired assessment may complete successfully; missing required evidence places this fixture's assessment execution in AwaitingEvidence. Apply Section II.D.a, not an implicit copy of the source date.

c) *Independent Resolution Branches:*

A — Passage of time. At 27 November 2026, 10:00 UTC, both output holdings are expired and ineligible for u,

although the original stock's accepted source cutoff has not passed. For either output holding c:

$$Q_{\text{recorded}}(c) = 50\,\mu L$$
$$Q_{\text{available}}(c, u, t) = 0\,\mu L$$

No movement, disposal, or holding closure is implied by expiration.

B — Missing condition evidence. Before either output cutoff, required storage evidence for C_B is missing. Its assessment is Unknown, not approval; available quantity for u is zero while recorded quantity remains 50 µL. A future accepted date alone cannot substitute for the missing evidence.

C — Amended source knowledge. Independently, at 25 November 2026, 12:00 UTC, authorized evidence changes the source cutoff to 25 November 2026, 10:00 UTC in A.expiry.v2. Preserve A.expiry.v1 and its authority, evidence, and versions. Trigger the reassessments required by A1 and B1, restricting use while required outcomes remain unresolved.

After authorized reassessment, C_A is expired under A1′s source cap. Under B1, the source was still stable at the recorded 09:00 preparation time; with the other conditions satisfied, C_B may retain its 26 November cutoff and remain valid at 12:00. Retain each new assessment's rule, evidence, source-version, and authority links. Neither source amendment nor reassessment changes quantities or silently rewrites descendant dates.

d) *Invalid Variations:*

- Expiration or missing stability evidence treated as material consumption or disposal.
- Missing assessment treated as approval, or Valid stability treated as unconditional eligibility.
- Aliquot handling history discarded because scientific identity is unchanged.
- A fresh stability period assigned automatically to every transformation.
- All descendants receiving a copied source date without policy-governed reassessment.
- Earlier assertions, governing versions, evidence, or authority overwritten.

L. Conformance Example #12 — Scientific Correction with Downstream Dependencies

a) *Scenario and Policy Fixture:*

Source holdings contain 100 µL each of distinct Materials A and B. A preparation execution records consumption of 80 µL A and 20 µL B to produce 100 µL of P, with composition assertion P.comp.v1 recording volume fractions 0.8 and 0.2. Assume additive volumes and no loss. Subsequent legitimate executions consume 20 µL P to produce 20 µL D and 10 µL P to produce 10 µL U, leaving 70 µL P. Each output retains its source and generating-execution references.

Pinned Specifications define this fixture's dependencies: D's calculated concentration assertion and use approval reference P.comp.v1 and require reassessment when it changes. U's approval instead depends on an independent accepted measurement; its Specification does not require reassessment for this source-composition change. P's identity rule permits the corrected proportions under the same identity; identity-changing corrections would require a separate authorized identity assessment, not silent relabeling.

b) *Correction and Execution Contract:*

New evidence establishes that the preparation actually consumed 70 µL A and 30 µL B. Recording that evidence alone changes neither quantities nor approval authority. An authorized correction execution binds the governing versions, original movements, assertions, and evidence. Its checkpoints establish the actual inputs, current-state accounting validity, scientific identity decision, superseding assertion, and protection against stale approval before correction completion.

Replace the two erroneous input-consumption movements with 70 µL A and 30 µL B using validated reversal/replacement groups. The net corrections return 10 µL to A's holding and debit another 10 µL from B's holding, with balancing consumption-position postings. Source balances become 30 µL A and 70 µL B. Preserve P's 100 µL production and both downstream consumptions; P remains at 70 µL, D at 20 µL, and U at 10 µL. Commit authorized P.comp.v2 with fractions 0.7 and 0.3, explicitly superseding P.comp.v1 while preserving both versions and their evidence.

Do not assume a shared transaction across scientific and decision authorities. Whatever coordination is used must enforce Section VII's stale-approval protection from the relevant scientific correction's commitment.

c) *Expected Results and Invalid Variations:*

- D's affected assertion and approval require reassessment. While mandatory reassessment is unresolved, a downstream task cannot use the prior approval; eligibility for that use is not approved and available quantity is zero, while recorded D quantity remains 20 µL.
- Test a use request after correction commitment but before dependency processing completes, including a request prechecked before correction. Neither may rely on the stale approval to authorize use.
- Authorized reassessment reaffirms, amends, or restricts D under its Specification, retaining prior assertions and decisions and recording the new source-version, evidence, and authority links. Missing evidence leaves reassessment unresolved.
- U's unaffected approval remains usable if its other eligibility conditions hold. Shared ancestry alone must not invalidate it.
- Reassessment propagates further where required by dependent Specifications, not through indiscriminate descendant invalidation.
- Reject correction groups that violate current quantity or other invariants; do not erase downstream consumption, rewrite observations, fabricate stock, or retarget historical Material references to force acceptance. No correction or reassessment implies a new physical preparation.

d) *Active Workflow During Scientific Correction:*

This branch starts after D and U are produced but before the correction above commits. A workflow W admits 20 µL of D for an assay requiring D's accepted composition-depen-

dent authorization. Pre-flight eligibility is satisfied under the original accepted assertion P.comp.v1. The workflow records the relevant authorization references and establishes custody according to its external allocation contract. Admission and custody alone leave D's recorded quantity unchanged.

W authorizes and initiates a robotic transfer using D. Before the instrument confirms the transfer outcome, the correction execution commits P.comp.v2, superseding P.comp.v1. D's governing Specification requires reassessment of the affected authorization.

The instrument may finish the operation already initiated. Its confirmed or unresolved physical outcome is handled using the existing TaskExecution and MaterialMovement rules. No physical reversal is inferred from the scientific correction. Confirmed transfer postings update the actual source and destination holdings; the earlier 20 µL balance for D is not assumed to remain in its original holding after transfer.

Before W initiates its next composition-dependent operation, the workflow must satisfy its continuation conditions or obtain an explicit intervention decision from an operator with the required authority. The decision must retain the corrected evidence, governing Specification, permitted exception or resolution, and effective history.

Expected results:

- Pre-flight authorization and the subsequent correction remain traceable.
- The operation already underway is resolved according to physical evidence, not retroactively canceled. Confirmed effects remain recordable after correction; uncertain or partial outcomes follow the existing execution rules.
- The next affected operation cannot begin using the stale authorization, even before a delayed correction notification arrives. A prechecked but not yet initiated operation must pass the applicable authorization boundary under the corrected state.
- An unaffected operation may proceed if its governing rules permit independent execution.
- Authorized continuation, rework, quarantine, or termination preserves the earlier committed movements and execution history. Continuation requires an applicable permitted exception or resolution, never waiver of a non-waivable requirement.
- A duplicate or delayed correction notification cannot bypass or duplicate the required intervention.
- An operator lacking the required authority cannot permit continuation.

X. Validation and Implementation Mapping

A complete executable conformance case must specify:

1) Natural-language scenario.
2) Initial state.
3) TaskSchema and mandatory lifecycle contract.
4) Relevant operation composition or dependencies.
5) Execution events and checkpoint evaluations.
6) Lifecycle transitions.
7) Proposed and committed effects.
8) Expected resulting state.
9) Applicable invariants.
10) Invalid variations and failure behavior.

The twelve worked cases above establish core semantic expectations; not every case yet supplies a complete executable fixture. The coverage inventory below includes both worked scenarios and scenarios to develop under this methodology:

Scenario	Primary modeling concern
Aliquot a sample	Shared identity, transfers, interruption
Transfer into a plate well	Unified Container model, atomic content creation
Move a vial between boxes	Placement constraints
Store or process material at a qualified location	Versioned storage conditions, containment gates, process capabilities, and capacity; Section II.D, Section III.E.a, and Section VIII.E
Consume a reagent	Quantity accounting
Dilute a sample	Identity and quantity transformation
Digest a protein	Scientific derivation and irreversible consumption
Mix a preparation	Composition without duplicate inventory
Independently produce protein batches	Material identity boundaries
Dispose of material and retire container	Ordered dependency and transaction boundaries
Correct concentration metadata	Scientific correction without physical movement
Correct a preparation after downstream use	Assertion-version dependencies, preserved accounting, and reassessment without stale authorization; case 12
Reverse an erroneous transfer	Historical correction
Replace an incorrect transfer	Supersession and atomic correction
Reconcile stock	Recorded versus observed quantity
Place material on QC hold	Decision state versus material state
Provenance-dependent expiration	Stability, holding history, and policy-triggered reassessment without quantity movement
Release material for specific use	Context-dependent eligibility
Admit a workflow and intervene after correction	Pre-flight eligibility, continuation boundaries, and operator authority; Section VIII.E and case 12
Release analytical results (interface)	Delegation to result-domain authority
Automated transfer timeout	Uncertain execution, safe retries, and effect commitment

Conformance review checks both directions: every required invariant needs a test, and every test outcome must follow from a definition or guarantee. Material-model revisions follow the same derivation method (Foundations, Section VII). Additional domain constraints must remain consistent with the pinned foundational contract; a missing material scenario is not by itself a reason to redefine the algebra.

A. Two-Level Implementation Evidence

The foundational obligations F-01 through F-12 in Foundations, Section VIII require independent evidence. The material cases add the following implementation mappings.

Domain concern	Required mapping and evidence
Identity and Specifications	Stable identifiers, pinned Specification versions, scientific validation, and amendment history; identity preservation and separation in cases 1 and 7
Containment and placement	Stable holding identifiers, active holding uniqueness, Material/Container links, fixed positions, compatibility, and acyclicity checks; well constraints in case 2
Quantities and movements	Material/dimension balance, individually identified postings without deduplication, compatible units, nonnegative active quantities, atomic commitment, and correction integrity after later activity; cases 1–5 and 8–10
Provenance and composition	Source and execution references, contribution values and measurement basis, versioned scientific dependencies, and absence of duplicate stock; cases 3, 8, 11, and 12
Decisions and eligibility	Decision scope, evidence, authority, effective history, intended-use assessment, versioned stability assertions, policy-triggered reassessment, and derived availability without changing recorded quantity; cases 6, 11, and 12; admission evidence and no new authorization on stale approval after a relevant correction commits
Execution and recovery	TaskSchema versions, lifecycle transitions, checkpoint evidence, effect identities, dependencies, uncertainty, and safe retries; cases 9 and 10. Workflow admission and continuation checks, enforceable restriction/version barriers, scoped operator authority, and duplicate/delayed notifications; Section VIII.E and case 12

The table identifies traceability obligations, not mandatory database tables or service boundaries. Every proposed schema element or architectural responsibility must link to a domain concept, owning authority, invariant, and conformance test. Additional fixtures are needed for the coverage inventory above, especially concurrency, relocation, containment cycles, and unauthorized operations. Include distinct postings with equal payloads, repeated same-Material receipts into one Container, availability under Unknown eligibility, and correction after legitimate subsequent consumption.

Location conformance evidence must additionally exercise nearest-declaring-location storage lookup, missing condition history, and relocation between qualified and unqualified storage. Under a fixture policy allowing six months at a specified −80 °C range, calculate stability from qualifying history rather than a current freezer label. Test a BSL-2-required use outside its authorized scope (No), missing containment evidence (Unknown), a characterization step at an incompatible location, and capacity overflow. None of these assessments independently changes recorded quantity.

B. Review Deliverables

A material implementation provides a derivation map, a correctness contract, and conformance evidence. Together they must demonstrate scientific identity, provenance, containment, quantity correctness, historical integrity, operational eligibility, and execution correctness.

A domain-specific design choice is acceptable only if it satisfies both this specification and the pinned foundations. Future specifications for instruments, results, quality, or scheduling can specialize the same foundations without importing this material-accounting model.